

A Mini-Project 1 Report on

Lightweight Anomaly Detection for Wearable Data

Using Autoencoders

By

K. Keerthana (24251A05L9)

M. Divya Sree (24251A05M9)

P. Sai Priya Reddy (24251A05R3)

under the guidance of

Dr. A. Sharada
Professor & Head
Department of CSE



Department of Computer Science & Engineering
G. Narayanamma Institute of Technology & Science
AUTONOMOUS (For Women)
Shaikpet, Hyderabad 500 104.

Affiliated to
Jawaharlal Nehru Technological University Hyderabad
Hyderabad – 500 085
June, 2026

A Mini-Project 1 Report on

Lightweight Anomaly Detection for Wearable Data

Using Autoencoders

**Submitted to the Department of Computer Science & Engineering, GNITS in the
partial fulfillment of the academic requirement for the award of B.Tech (CSE)
under JNTUH**

By

K. Keerthana (24251A05L9)
M. Divya Sree (24251A05M9)
P. Sai Priya Reddy (24251A05R3)

under the guidance of

Dr. A. Sharada
Professor & Head
Department of CSE



Department of Computer Science & Engineering
G. Narayanamma Institute of Technology & Science
AUTONOMOUS (For Women)
Shaikpet, Hyderabad- 500 104.

Affiliated to
Jawaharlal Nehru Technological University Hyderabad
Hyderabad – 500 085
June, 2026

G. Narayanamma Institute of Technology & Science
AUTONOMOUS (For Women)
Shaikpet, Hyderabad – 500 104.
Department of Computer Science & Engineering



Certificate

This is to certify that the Mini-Project 1 report on “Lightweight Anomaly Detection” for Wearable Data Using Autoencoders is a bonafide work carried out by K. Keerthana (24251A05L9), M. Divya Sree (24251A05M9), P. Sai Priya Reddy (24251A05R3) in the partial fulfillment for the award of B.Tech degree in Computer Science & Engineering, G. Narayanamma Institute of Technology & Science, Shaikpet, Hyderabad, affiliated to Jawaharlal Nehru Technological University, Hyderabad under our guidance and supervision.

The results embodied in the project work have not been submitted to any other University or Institute for the award of any degree or diploma.

Internal Guide
Dr. A. Sharada
Professor & Head
Department of CSE

Head of the Department
Dr. A. Sharada
Professor & Head
Department of CSE

Acknowledgements

We would like to express our sincere thanks to **Dr. K. Ramesh Reddy**, Principal, GNITS, for providing the working facilities in the college.

Our sincere thanks and gratitude to **Dr. M. Seetha**, Professor CSE, Dean R&D, **Dr. N. Kalyani**, Professor CSE, Dean I&I, GNITS, for all the timely support and valuable suggestions during the period of our project.

We extend our heartfelt gratitude to **Dr. A. Sharada**, Professor & Head, Department of Computer Science and Engineering, GNITS, for her unwavering support and invaluable guidance throughout our project, providing timely assistance and insightful suggestions.

We are extremely thankful to **Dr. P. Sunitha Devi**, Assistant Professor Mini-Project 1 overall coordinator, Department of CSE, GNITS for all the valuable suggestions and guidance during the period of our project.

We are also extremely thankful to our project coordinators **Dr. T. Divya Kumari**, Assistant Professor and **Mrs. V. Subha Lakshmi**, Assistant Professor, Dept. of CSE, GNITS, for their encouragement and support throughout the project.

We are extremely thankful and indebted to our internal guide, **Dr. A. Sharada, Professor & Head**, Department of CSE, GNITS, for her constant guidance, encouragement, and moral support throughout the project.

Finally, we would also like to thank all the faculty and staff of CSE Department who helped us directly or indirectly, parents and friends for their cooperation in completing the project work.

K. Keerthana (24251A05L9)

M. Divya Sree (24251A05M9)

P. Sai Priya Reddy (24251A05R3)

G. Narayanamma Institute of Technology & Science
AUTONOMOUS (For Women)
Shaikpet, Hyderabad – 500 104.

Department of Computer Science & Engineering

COE for Advanced Research in Artificial Intelligence



Certificate

This is to certify that K. Keerthana (24251A05L9), M. Divya Sree (24251A05M9), P. Sai Priya Reddy (24251A05R3) II B.Tech, has successfully completed project work in “COE for **Advanced Research in Artificial Intelligence**”, CSE Department, as part of ongoing research activities carried out in COE.

The project titled “Lightweight Anomaly Detection for Wearable Data Using Autoencoders” that is being submitted in partial fulfillment for the award of B.Tech, Computer Science and Engineering, G.Narayanamma Institute of Technology & Science affiliated to JNTUH, Hyderabad is a record of bonafide work carried out by them in our guidance and supervision.

Internal Guide

Dr. A. Sharada,
Professor and Head,
Department of CSE

CoE Incharge

Dr D.V. Lalitha Parameswari
Associate Professor,
Department of CSE.

Head of the Department
Dr. A. Sharada
Professor and Head
Department of CSE

ABSTRACT

The growing popularity of wearables, however, presents an opportunity to constantly monitor physical and physiological signals, which may lead to the identification of circadian changes earlier than usual. However, detecting such deviations is hard due to unstructured and noisy nature of the sensor data and insufficient computing capabilities of mobile hardware. In this study, a lightweight anomaly detection method for wearable data based on autoencoders will be proposed. Special emphasis will be paid to the structured pre-processing of signals to filter out noise without adding extra overhead.

To detect anomalies in a user's circadian rhythm, the system uses a small autoencoder trained on data from healthy individuals. It identifies unusual patterns by measuring the reconstruction error, which is the difference between actual and predicted data. The model was evaluated for accuracy and speed, showing reliable results while using very little computational power. The system demonstrates that a simple machine learning model can effectively detect anomalies without requiring labeled data. By learning an individual's normal behavior, it can identify changes in their circadian cycle.

Keywords: Detecting Anomalies, Autoencoder Model, Wearable Data, Circadian Rhythm Disruptions, Digital Health, Visualization Dashboard, Lightweight model, Deep Learning

SDGs on Sustainability:

Goal 3: Quality Health and Wellbeing – It supports the use of technologies which allows early detection of circadian disruption and also provides for ongoing monitoring and support for health using wearable technologies.

Goal 9: Industry, Innovation and Infrastructure – It develops efficient lightweight AI models that are appropriate to use in resource constrained and edge computing environments.

Table of Contents

Sl.No.	Topic	Page No.
	Abstract	v
	List of Figures	vii
	List of Tables	viii
1.	Introduction	1
	1.1 The Study of Circadian Rhythm Disruptions	1
	1.2 Problem Statement	4
	1.3 Existing Solutions	5
	1.4 Drawbacks of Existing System	6
	1.5 Proposed System	7
	1.6 Objectives	9
	1.7 Methodology	9
	1.8 Organization of the Project	11
2.	Literature Survey	12
3.	Lightweight Anomaly Detection	16
	3.1 Architecture of the System	16
	3.2 Description of Modules	17
4.	Implementation	25
	4.1 Datasets	25
	4.1.1 Data processing and Normalization	25
	4.1.2 Training the Autoencoder Model	26
	4.2 Technologies used	28
	4.3 Autoencoder Model	30
	4.4 Statistical Method	32
	4.5 Comparisons of Algorithms	34
5.	Results and Discussions	36
	5.1 Results	36
	5.2 Performance Measures	41
	5.3 Comparative Analysis of Results (Tables and Graphs)	42
6.	Conclusions and Future Enhancements	46
	References	47
	Glossary	48
	Appendix	50

List of Figures

Fig No.	Figure Name	Page no
1.1	Wearable Devices	1
1.2	Methodology	10
3.1	Architecture of the System	16
4.1	Architecture of autoencoder	27
5.1	Login Module for Secure Access	36
5.2	Date Selection with Dashboard	37
5.3	Anomaly Detection Graph	37
5.4	Stress Analysis	38
5.5	Heart Rate Monitoring	39
5.6	Manual Health Input	39
5.7	Daily Summary and Insights	40
5.8	Statistical anomaly detection using mean and standard deviation thresholds	43
5.9	Autoencoder-based anomaly detection using reconstruction error	43

List of Tables

Table No.	Table Name	Page no
1.1	Common Wearable Data Parameters	2
1.2	Effects of Circadian Disruption	3
1.3	Problem Analysis	4
1.4	Comparison of Existing Methods	6
5.1	Comparison of Algorithms Results	42

1. INTRODUCTION

1.1 The Study of Circadian Rhythm Disruptions

Wearable technology has found its way into the healthcare sector as a vital component of the contemporary healthcare system because of its ability to provide constant access to human physiological and activity-related statistics. Smart devices like smart watches capture real time data like heart rate, number of steps, and sleep patterns, which record how a person behaves in everyday life. Wearable devices, unlike more traditional ways of healthcare which are based on infrequent clinical observations, offer a constant flow of information, so it is now possible to assess long-term patterns, and even subtle behavioral shifts. The increasing availability of such data has created new opportunities for early detection of health issues, particularly those related to lifestyle and daily routines. One of them is circadian rhythm, which controls the sleeping patterns, energy, and general body functioning.

1.1.1 Wearable Data and Its Significance:

Wearable data is a useful source of information on how people act during the day. This information is gathered on a continuous basis from different devices as showed in Figure 1.1 and this makes it more acceptable compared to manual health reporting. Through the analysis of this data, one can learn patterns, identify anomalies and give alerts or measures to take care of before things get out of hand.



Figure 1.1: Wearable Devices

- Real-time (Continuous monitoring) monitoring allows us for the observation of subtle changes in behaviors over time, instead of using the disconnected data that is not always relevant to the actual situation.
- Real-time data collection increases accuracy and eliminates reliance on a person to operate the system by employing real-time data collection, which is more effective in long-term health monitoring.
- There are many parameters which are collected from Wearable devices as shown in Table 1.1 and each of them have their own importance and play a major role in the healthcare.

Table 1.1: Common Wearable Data Parameters

Parameter	Description	Importance
Heart Rate	Beats per minute	Stress and cardiovascular health
Step Count	Monitoring the movement	Measures the level of physical activity
Energy Burned	Calories burned	Demonstrates metabolic activity
Sleep Duration	Analyses sleep hours	Helps in decoding sleep patterns
Respiratory Rate	Breathing cycles	Indicates respiratory health

1.1.2 The Circadian Rhythm

The circadian rhythm is a biological cycle that occurs over 24 hours. This process manages how the body functions, how it sleeps, uses energy from food and releases chemicals. If this rhythm is consistent, then the body works in a way that is effective and keeps a state where activity plus rest is equal.

- It controls when a person is awake or asleep and helps energy levels stay at the same state during the day.
- As a result, the body performs physical and mental tasks at a level that is expected for normal function.
- By its action, it affects the balance of chemicals but also how the body processes nutrients.
- Those functions are necessary because they keep the body in a state of health and stop illnesses that last for a long time.

1.1.3. Results when the Circadian Rhythm is Disrupted

Disruption to the circadian rhythm happens when daily habits are not regular. For many people this occurs because of changes in how they live or because of things outside the body. The effects of Circadian Disruption are listed below in Table 1.2.

When those disruptions continue, they cause problems for the body and the mind.

- If sleep happens at different times, a person feels tired as well as is less able to focus.
- To be productive or to think clearly becomes difficult under the conditions.
- In the long term, those disruptions make it more likely that a person develops conditions like heart disease or high body weight.
- And there is a higher chance that a person can experience disorders that are related to tension.

Table 1.2: Effects of Circadian Disruption

Area	Impact
Sleep	Insomnia, irregular sleep patterns
Mental Health	Stress, anxiety, decreased attention
Physical Health	Heart diseases, obesity
Productivity	Loss of efficiency and exhaustion.

1.1.4 Requirement for Analysis via Intelligence

Although wearable devices create data in high quantities, the extraction of information that contains meaning is difficult. The data is frequently inconsistent and lacks certain values and it changes in a large degree among different people. On account of those factors, standard ways to analyse the data do not produce results.

- To process sets of data that occur over time plus are large, researchers use techniques for analysis that are advanced so they can find patterns which are important.
- It is necessary for the system to change based on how a person acts, because every user possesses physical attributes and habits that are distinct.

1.2 Problem Statement

As people use wearable devices more often, the devices produce continuous data in high volumes but the process to analyse this data in a way that is fast and correct is a primary difficulty. If we look at current systems, they depend on limits that are set or models that are complex but also both options have flaws during use in real time. The major challenges faced when working on this problem are listed in Table 1.3.

Challenges that are Primary

- **Lack of personal adaptation**

By using current systems, the same rules apply to every person, which means the systems disregard how habits and physical patterns are different for each individual. Because of this the systems make predictions that are not correct and the ability to find data that is unusual is lower.

- **Degree of difficulty in wearable data**

Because wearable data depends on time as well as changes constantly, it contains signals that are unwanted and values that are absent. In the situations, old ways of working fail to manage such a high degree of difficulty.

- **Limits on computation**

For many models that are advanced, the requirement for power to process data is high. And for this reason, they are not appropriate for use on devices that are wearable.

Table 1.3: Problem Analysis

Challenge	Description	Impact
Personalization	Same model for all users	Low accuracy
Data Complexity	Time-series, noisy data	Hard to analyse
Computation	High processing requirement	Not real-time

Core Objective

The core aim of the proposed project is to design an anomaly detection system that will be able to analyse wearable data, identify normal patterns, and detect circadian rhythm disturbances.

1.3 Existing Solutions

Several approaches have been created to analyse wearable data and detect anomalies. The approaches vary in terms of complexity, accuracy, and applicability to the real-life scenarios. However, there are few limitations in existing solutions as shown in the Table 1.4 which we want to overcome in our project and give the best result.

Rule-Based Approaches

- These approaches utilize some predetermined thresholds for classification into 'normal' and 'abnormal'. This approach is quite simplistic yet not flexible enough.
- Because the same set of rules applies to everyone, it cannot accurately analyse real life dynamic data.

Statistical Approaches

- These approaches rely on deviations from mean values and therefore can easily deal with simple cases but not with complicated data.
- This approach does not account for time dependences and is prone to noise in data.

Machine Learning Approaches

- Machine learning approaches have the ability to analyse multi-dimensional data and discover hidden patterns within.
- However, machine learning requires adequate training and computing resources.

Systems Based on Autoencoders

- These algorithms can learn normal behaviour and spot any anomalies by comparing reconstruction errors, hence making them good candidates for wearables' time series data analysis.
- They do not need labelled data, which makes them useful in real-world situations, where labelling of data may prove to be very challenging.

Table 1.4: Comparison of Existing Methods

Method	Advantage	Limitation
Rule-Based	Simple	Not adaptive
Statistical	Fast	Not accurate
ML Models	Powerful	High computation
Autoencoder	Learns patterns	Needs optimization

In light of the above discussion, it is evident that current methods lack the ability to offer an effective and efficient way of dealing with all the aspects associated with the problem at hand. Hence, there is a need for an effective system to solve the issue at hand.

1.4 Drawbacks of Existing Systems

Anomaly detection systems currently used for wearable data analysis employ conventional statistics or more sophisticated machine learning algorithms. Nonetheless, both types of approaches suffer from several important problems:

- **Computational Expensiveness:** The use of deep neural networks and other highly complicated models requires a lot of computational resources. These models cannot be implemented in wearable technology due to the low computing capacity and energy efficiency limitations inherent in small wearable devices.
- **Imperfections in Handling Imperfect Data:** The raw sensor data collected from wearables can contain many imperfections such as noise, missing values, or inconsistencies, which are caused by device-related restrictions and users' activities. Most conventional models cannot handle such noisy data efficiently, resulting in reduced precision of anomaly detection.
- **Dependence on Labelled Data:** Some machine learning models depend significantly on labelled datasets, which can be obtained through tedious, costly, and laborious data preprocessing conducted by experienced practitioners.

Thus, this method is not scalable in practice and cannot effectively identify uncommon anomalies.

- **Offline-Based Nature of the Analysis:** Many contemporary anomaly detection systems do not operate in real time but rather work offline, processing data sets at once. In such case, the process is time-consuming, delaying anomaly identification.
- **Not Ideal for Edge Computing:** Anomaly detection models usually are computationally expensive and require strong computing capability to run. Wearables and edge devices have limited computational capabilities, making it challenging to apply such models effectively. This limits the usage of such models in practice where efficiency and energy savings play a critical role. The existing models and methods do not efficiently run on edge and wearables for practical use. There is therefore a need for an efficient system for real-time wearable devices

1.5 Proposed System

The suggested solution can be seen as an intelligent way of analysing smartwatch data by using machine learning and explainable AI approaches. The system will offer valuable insights instead of only providing data. The description of the main components of the proposed system is provided below:

Key Components of the Proposed System

- **Real-Time Smartwatch Data Integration**
The suggested system will operate using data obtained from a smartwatch and related to the heart rate, number of steps taken, and energy spent. Such data reflect the actual activity of people, as demonstrated in the dashboard outputs.
- **Specialized Preprocessing Algorithm**
The system includes a customized preprocessing algorithm that cleans and formats the obtained data into a suitable structure. It helps prepare the input data for the further analysis process.

- **Feature Engineering**

Additional features are introduced into the system through feature engineering. Such features reflect the activity of people at different moments, making it easier for the model to detect any behaviours people.

- **Machine Learning-based Anomaly Detection**

The system uses the anomaly detection algorithm to identify any abnormalities in people's behaviour. Anomalies are detected by analysing graphs.

- **Model Implementation and Management**

The machine learning algorithm model is executed via a specific module. Moreover, the model can be trained via normal behaviour patterns and be used to evaluate the entered information to ensure that proper anomaly detection is done. The model uses unsupervised learning algorithms (autoencoder) to reduce dependency on labelled datasets.

- **Explanation Module**

Another module has been incorporated into the system to make its workings easier to comprehend by users. This explanation module explains why certain data points are recognized as anomalies in the dataset.

- **Streamlit Visual Dashboard**

A visualization dashboard has been developed for the system outputs via Streamlit. With this in mind, the following visual elements have been integrated:

1. Heart rate graph
2. Steps statistics
3. Burnt active energy graph
4. Anomalies detection graph

Such visual elements enable users to gain insight into their health information and detect any anomalies.

- **Minimal Time for Processing Collected Data**

The software will ensure that the data collected is processed within a very minimal period to allow real-time information on the dashboard.

- **Scalability by Having Different Project Modules**

The whole project will be divided into various modules such as auth, app, data, and dashboard modules; this helps in scaling because any further features and models can easily be added to the project without disrupting the other portions of the project.

- **The Output Will Be Presented in a Simple Way**

The last point in the project design is the easy-to-understand format of presenting the results, and the output will include numbers and graphs.

1.6 Objectives

- To collect and preprocess wearable sensor data for accurate and efficient analysis of user health patterns.
- To design and develop a lightweight autoencoder model that learns normal physiological behaviour.
- To detect anomalies in health data that may indicate disruptions in circadian rhythm.
- To enable real-time monitoring with minimal computational cost for practical deployment.
- To present results through an interactive system that provides clear insights into user health and anomalies.

1.7 Methodology

The design process as shown in Fig. 1.2 will be implemented in a systematic and logical manner to ensure that the anomaly detection tool based on the health data is created efficiently.

- **Data Collection**

Heart rate and number of steps together with timestamps are collected through an Apple Smartwatch device while users perform various kinds of activity.

- **Data pre-processing**

The gathered data will undergo data cleaning in which missing values and inconsistencies will be handled. Also, timestamps will be converted and re-sampled using an hourly timeframe and applying feature selection.

- **Feature Engineering**

Pre-processed data will undergo feature engineering to be normalized and transformed. Sequences of time series will be created via a sliding window technique, and a threshold value (95%) will be established.

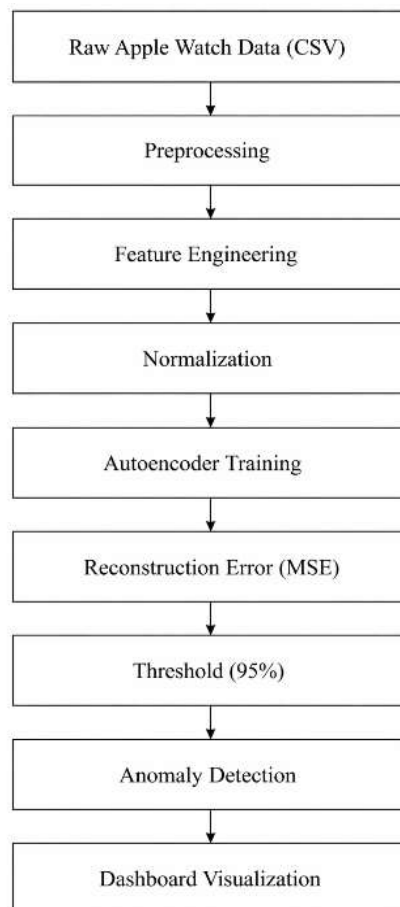


Figure 1.2: Methodology

- **Model Development**

A simple autoencoder will be developed for the detection of anomalous patterns by training only the normal patterns in the dataset.

- **Anomaly Detection**

This process will identify anomalies by comparing the inputted data with the output from the model. All data points with more than the set threshold will be considered anomalies.

- **Model Evaluation**

To determine the capability of the developed model to detect anomalies, performance will be analysed based on reconstruction error patterns.

- **Visualization and Dashboard**

Graphical visualization of results in the form of line charts and others will be presented via Streamlit dashboard.

1.8 Organisation of the Project

This project report contains several chapters in order to present a detailed view of the developed anomaly detection system for wearable health devices.

Chapter 1: Introduction to the project, which includes goals, motivations, problem statements, scope, and methodology. In addition, the importance of the system will be described.

Chapter 2: Literature survey on the development of anomaly detection techniques through statistical methods and other machine learning and deep learning-based approaches.

Chapter 3: Explanation of the system design and architecture, where all system components will be discussed in detail, starting from data acquisition and pre-processing until model processing and output visualization.

Chapter 4: Implementation process, which includes dataset preparation, pre-processing, feature extraction, model training, and parameter tuning. Moreover, the whole execution process of the system will be discussed.

Chapter 5: Presenting results and analysis, where anomaly detection performance will be illustrated through various graphical plots like line graphs.

Chapter 6: Conclusion part of the project, where important findings of the system performance will be presented. Besides, future improvements of the system will be suggested.

2. LITERATURE SURVEY

The evolution of wearable technology has significantly impacted modern healthcare systems, allowing for the continuous and immediate monitoring of physiological and behavioural data. Smart wearable devices, like fitness trackers and smartwatches, are gaining popularity as valuable devices used in health monitoring activities. Examples of popular devices include the Apple Watch, which is capable of monitoring several physiological data points, including heart rate, physical activity, sleep patterns, and energy expenditure.

Unlike current healthcare systems that focus on periodic check-ups, wearable devices offer continuous data collection opportunities and, therefore, enable prompt action in cases where anomalies can be detected early on. According to NIH, there is an increasing need for the adoption of wearable health monitoring devices within biomedical research, emphasizing the non-invasive and continuous data collection opportunities offered by these devices [6].

Additionally, scientific evidence provided through PubMed Central shows that wearable-based health monitoring systems can detect health anomalies in real time, especially in the management of cardiovascular and sleep disorders [10]. Finally, global bodies, such as the World Health Organization (WHO), recognize the importance of health monitoring in enhancing global healthcare systems, lowering costs associated with healthcare services, and providing solutions for managing chronic diseases [11].

The importance of understanding the processes taking place inside a person's body should be emphasized in terms of anomalies detection within health data. In particular, the human behaviour is dictated primarily by internal biological processes. One of the most important aspects of human physiology is the circadian rhythm, which is known to have a great effect on one's sleep-wake cycle.

As stated by the National Library of Medicine, disruption of the circadian rhythm may cause serious consequences including sleep problems, fatigue, heart diseases, and some mental disorders [7]. Hence, analysis of circadian rhythms becomes very important for detecting the deviations in patients' condition. Moreover, according to scientifically proven guidelines issued by the National Sleep Foundation, people need to have sufficient sleep of high quality depending on their age [8]. This means that the norms concerning the amount of sleep a person should get during night time should be taken into account. With regard to the above facts, one can note that the discussed project will benefit greatly from incorporating information about circadian rhythms and sleep norms.

Feature selection is another critical consideration when developing an effective wearable data analysis model. With wearable sensors producing tremendous amounts of data based on different parameters, it is important to select which among these data can contribute more significantly to detecting anomalies in data behaviours. According to a study published in the MDPI Sensors Journal, machine learning algorithms can be effectively applied in the analysis of wearable sensor data.

This paper highlights the importance of selecting significant physiological features such as heart rate, step count, and heart rate variability (HRV) [5]. These physiological parameters provide crucial data related to one's level of physical activity and his or her health status. It is therefore expected that choosing these as data parameters would allow the detection of anomalies in data behaviour more effectively.

Data preprocessing is an important aspect in the machine learning process because the quality of input data plays a key role in improving the efficiency of the algorithm. Data acquired from wearable sensors tends to be inaccurate and incomplete due to different limitations. Geron [1] argues that appropriate data preprocessing is crucial for designing high-quality machine learning algorithms. As part of this project, the following preprocessing steps will be performed.

First, timestamps will be transformed to datetime format. Then, the data will be sorted chronologically, missing values will be appropriately handled, and all features will be converted to numeric format. Moreover, resampling methods will be employed to adjust time intervals. Specifically, the time series data will be transformed to an hourly format in which heart rate values will be averaged, while step counts will be summed up.

Another preprocessing technique that is important is normalization, which ensures that all of the features play an equal role while building the model. As wearable data contains features in different ranges, such as heart rates and steps, normalization ensures that one particular feature does not overpower other features. According to Geron [1], there are various methods for normalization, such as MinMax scaling, which rescales data between two predefined limits, usually zero to one. This process is used because it enhances the stability of the algorithm, as well as its convergence.

Feature engineering is also done in order to improve the effectiveness of the dataset, which means deriving more variables from existing data. For example, activity-level and night indicators could be used as features. According to the findings of the research published in the PLOS ONE journal, machine learning techniques should be used to analyse time-series data in order to detect anomalies in the behaviour of users [9].

The main technique utilized for this research project is an autoencoder. It is an unsupervised deep learning method used in training machine learning models. Autoencoders aim at obtaining a compressed representation of input data by first encoding it and later decoding it to reproduce the original data. This mechanism allows the model to discover hidden patterns in the dataset.

The fundamental principles of deep learning and neural networks are covered in detail by Goodfellow et al. [4], who discuss neural network architectures, training methods, and optimization. On the other hand, Geron [1] focuses on providing practical approaches to building autoencoders using frameworks like TensorFlow and Scikit-learn. They play a vital role in building the model applied in this research. Through the use of autoencoder, normal behaviours can be accurately reconstructed while abnormal behaviours cannot be decoded successfully.

The technique used in detecting anomaly in the proposed framework is that of reconstruction error, which is an indication of the difference between the input data and the output data produced by the autoencoder algorithm. The theoretical underpinning of the method used to detect anomaly is based on outliers, which is a concept described by Aggarwal [2] as any point in the dataset which is far removed from the norm in the particular data set.

Reconstruction error in this case is measured using the mean squared error (MSE) measure, which indicates the level of deviation in the dataset between the input and the output data. Anomalies will then be detected by setting a threshold limit at the 95th percentile of the reconstruction error [2].

Anomaly interpretation needs an awareness of the physiological standards and health guidelines. The CDC provides some reliable criteria regarding the normal heart rates and levels of physical activity, which are used to assess the health condition [3]. Comparing detected anomalies to the established physiological standard is a good way to get useful information about possible health concerns.

For instance, abnormally high heart rate may be due to stress or heart-related problems, while abnormally low activity may signal sedentarism or tiredness. In addition, disrupted sleep pattern is another criterion for anomaly detection. The use of such guidelines contributes to more effective interpretation of anomalies.

In summary, it is evident that the reviewed literature indicates the successful integration of wearable devices, biology knowledge, and machine learning algorithms for anomaly detection purposes. The process involves collecting information on a constant basis and then applying sophisticated pre-processing techniques. It further requires feature extraction from such data, developing complex neural networks, and interpreting physiological information.

Every study cited in this survey plays an integral role in different stages of the project implementation. Overall, the reviewed literature provides the necessary scientific foundation for building a scalable, efficient, and accurate model for anomaly detection and prevention.

3. Lightweight Anomaly Detection

3.1 Architecture of the System

The architectural design as shown in the Fig. 3.1 of the presented system includes the development of a well-designed framework for the pipeline that can convert the raw data collected using wearables to insight through the following processes:

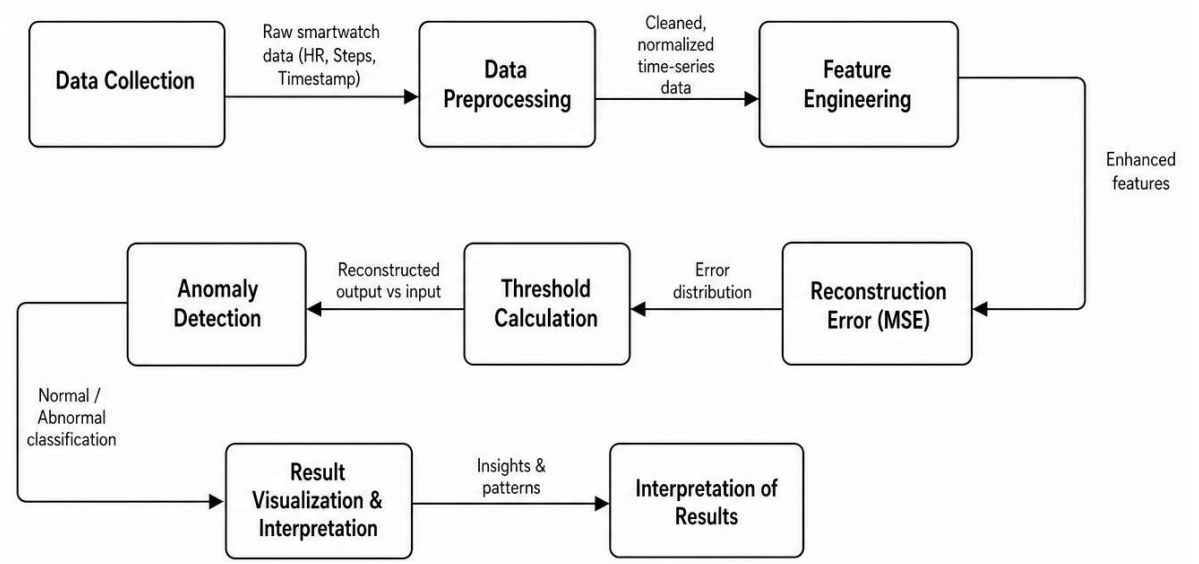


Figure 3.1: Architecture of the System

- **Wearable Data Collection (Apple Smartwatch)**

To start with, our proposed system will use the smartwatch of Apple Inc. to collect real-time physiological and activity data. The important parameters of interest, such as the heartbeat rate, the number of steps, and timestamps, will be collected in line with the daily activities of the user.

- **Data Preprocessing**

Once the data has been acquired, it will undergo preprocessing whereby missing data will be removed or imputed. Also, data preprocessing entails filtering out noise in the collected data. Lastly, data preprocessing includes converting timestamps to a common format.

- **Feature Engineering**

At this step, the cleaned data will be converted into a structured time series sequence where temporal dependencies are captured.

- **Machine Learning Algorithm Based on an Autoencoder Model**

A simple machine learning algorithm in form of autoencoder model will be used to learn the regular activity pattern of the user.

- **Error Calculation of Autoencoder Model**

The error associated with an autoencoder model will be calculated after applying it to preprocessed data.

- **Anomaly Detection Process**

Anomalies in data are detected by comparing them against a certain threshold value based on the reconstruction error. Any data point with errors greater than the set threshold is considered anomalous, implying some unusual behavior from the users.

- **Results Visualization via Interactive Dashboard**

The visualization of the results is done through an interactive dashboard that displays the data in the form of graphs and charts, aiding the interpretation of activities, outputs, and anomalies. Results visualization is done through an interactive dashboard created using Streamlit.

3.2 Description of Modules

The proposed system is comprised of various modules which undertake different operations in the pipeline. All these modules have been interconnected with one another to facilitate efficient transformation of the raw data captured by the smartwatch into information through visual presentation.

The system consists of the following modules:

- i. Data Collection Module
- ii. Preprocessing Module
- iii. Feature Engineering Module
- iv. Data Sequencing for Wearable Data Module
- v. Autoencoder Model Development Module
- vi. Reconstruction Error (MSE) Calculation Module

- vii. Threshold Calculation Module
- viii. Anomaly Detection Module
- ix. Results Interpretation Module

3.2.1 Data Collection Module

This module collects data regarding physiological and behavioral activities from the Apple Smartwatch. The dataset will be acquired using an exported file format that includes time stamps. The main features included in this project are heart rate, number of steps taken, and timestamps. The features capture the behavior of the user and form the basis for processing the data. The module serves as the backbone of the entire system since anomalies cannot be detected without reliable data collection.

Direct Attributes taken from the Dataset:

- **Heart Rate**

The heart beats a number of times per minute. This is called the Heart Rate. It is measured in beats per minute or BPM. The Heart Rate shows how active the heart is.

- **Heart Rate Variability SDNN (HRV)**

The Heart Rate Variability SDNN measures how much time is between each heartbeat. It looks at how the heart beats and how much times between each beat. This shows if the heart is beating normally or not. The Heart Rate Variability SDNN also shows how well the body is handling stress.

- **Resting Heart Rate**

The heart rate when you are completely at rest shows how fit your heart is.

- **Step Count**

The Step Count is the number of steps taken. This shows how active the body is. The Step Count measures how many steps are taken in an amount of time.

- **Active Energy Burned**

The body uses energy when it is active. The Active Energy Burned is the amount of energy used when walking, running or exercising. This does not include the energy used when the body is at rest.

- **Basal Energy Burned**

The body always uses some energy even when it is at rest. The Basal Energy Burned is the amount of energy used by the body when it is not active. This energy is used for things like breathing and blood flow.

- **Physical Effort**

A normalized measure between 0 and 1 of how intense the physical activity undertaken by the user was.

- **Walking Speed**

The Walking Speed is how fast the body is walking. This is usually measured in kilometres per hour. The Walking Speed shows how mobile the body is. It also shows how intense the physical activity is.

- **Respiratory Rate**

It is the number of breaths taken per minute. This shows how the body is breathing. The Respiratory Rate can also show if the body is under stress or exerting itself.

3.2.2 Preprocessing Module

The data preprocessing module will help in cleaning, consistency, and standardization of collected data. Wearable data can have some missing value, improper timestamps, and noise in the data set. This problem can be fixed by filtering the data to remove irrelevant variables. Conversion of data to datetime and sorting the data on a chronological basis can solve the issue of timestamps. Resampling data at an interval of one hour helps to keep the data evenly spaced out in terms of time. Missing values can be imputed through the forward and backward fill technique. Normalization of data will be done through MinMax Scaling.

3.2.3 Feature Engineering Module

This is the section where more features are generated from the data that has been preprocessed. These features are: Stress Score, Sedentary Index, Recovery Index, Rolling Heart Rate, Rolling Step Count, and Rolling Energy Burned. They provide information related to stress, sedentary nature, recovery efficiency, and short-term trends.

i) Stress score:

Formula:

$$\text{Stress Score} = \frac{\text{HR}}{\text{HRV}}$$

Derivation:

The Stress Score derivation is based on physiological parameters. When an individual is stressed, the heart rate rises but heart rate variability drops because of the lack of flexibility in the body's autonomic system. Therefore, stress will be directly proportional to heart rate and inversely proportional to HRV. By bringing the two together, we arrive at a ratio formula.

ii) Sedentary index:

Formula:

$$\text{Sedentary Index} = \begin{cases} 1, & \text{if Steps} < 10 \\ 0, & \text{if Steps} \geq 10 \end{cases}$$

Derivation:

The derivation of the Sedentary Index feature utilizes a rules-based thresholding strategy. A very low number of steps is indicative of inactivity or sedentary behavior. The threshold is set at 10 steps/minute to determine whether a person is inactive or not.

iii) Recovery index:

Formula:

$$\text{Recovery Index} = \frac{\text{Active Energy}}{\text{HR} + 1}$$

Derivation:

Recovery Index uses the notion of physiological efficiency. An effectively recovered person will engage in physical activities with a comparatively lower heart rate. Hence, recovery is directly proportional to the energy consumed and inversely proportional to the heart rate.

iv) Rolling Heart Rate:

Formula:

$$\text{Rolling HR}_t = \frac{1}{N} \sum_{i=t-N+1}^t \text{HR}_i$$

Derivation:

Rolling Heart Rate is calculated by applying the rolling average method through time series analysis. The formula involves computing the average between the present value and the prior one.

v) Rolling Step Count:

Formula:

$$\text{Rolling Steps}_t = \frac{1}{N} \sum_{i=t-N+1}^t \text{Steps}_i$$

Derivation:

Rolling Step Count is determined based on a moving average to make sure that fluctuations in step counts are reduced. This is because step counts tend to vary greatly between different periods, but the moving average reduces the variance.

vi) Rolling Energy Burned:

Formula:

$$\text{Rolling Energy}_t = \frac{1}{N} \sum_{i=t-N+1}^t \text{Energy}_i$$

Derivation:

Rolling Energy Burned is estimated based on the moving average approach to help minimize fluctuations in energy burned. This ensures that the machine learning model can better learn from the overall trend in energy burned.

3.2.4 Data Sequencing for Wearable Data

To account for the presence of time dependency in the wearable data, the preprocessed data is structured into sequences employing a sliding window technique. By applying this technique, the model will have the capacity to learn time-related characteristics by analyzing continuous chunks of data.

Each chunk or sequence is made up of a time series that represents the temporal aspects of the physiological information captured through the sensors. For the purpose of this project, each sequence contains 12 time steps, which equates to 12 hours of data. The time steps contain engineered features developed at the feature engineering stage, which include the following:

- Stress Score
- Sedentary Index
- Recovery Index
- Rolling Heart Rate (HR_roll5)
- Rolling Steps (Steps_roll5)
- Rolling Energy (Energy_roll5)

Formula:

$$X = [x_1, x_2, x_3, \dots, x_L]$$

Attributes:

$X \rightarrow$ Sequence of inputs

$x_i \rightarrow$ Features vector at time step i

$L=12 \rightarrow$ Length of each sequence

$F=6 \rightarrow$ Number of Features

3.2.5 Developing Autoencoder model Module

The module for developing models uses an autoencoder neural network algorithm that learns the behavior patterns of the dataset without using any labels. This algorithm comprises two major parts called the encoder and the decoder. The encoder reduces the dimensions of the input to capture its essential information, which is then decoded by the decoder. Since the training is conducted only on the basis of normal samples, it is able to reconstruct normal behavior patterns and fails in reconstructing abnormal patterns.

Formula:

$$\hat{X} = g(f(X))$$

Attributes:

$X \rightarrow$ Original input

$f(X) \rightarrow$ Encoder

$g(f(X)) \rightarrow$ Decoder

$\hat{X} \rightarrow$ Output

3.2.6 Reconstruction Error (MSE)

The difference between the input data and the reconstructed data is computed through the mean squared error (MSE). This metric gives an indication on how successfully the model has been trained to recognize normal behavior; a low error implies successful reconstruction, whereas high errors represent abnormal data.

Equation:

$$MSE = \frac{1}{L \times F} \sum_{i=1}^L \sum_{j=1}^F (X_{ij} - \hat{X}_{ij})^2$$

Attributes:

$L \rightarrow$ Length of sequence

$F=4 \rightarrow$ Number of features

$X_{ij} \rightarrow$ actual value

$\hat{X}_{ij} \rightarrow$ Reconstructed value

$i \rightarrow$ Index of time series

$j \rightarrow$ Index of feature

3.2.7 Threshold Calculation

The threshold is the critical value used to separate the normal and anomalous data. The threshold value in this study is taken to be the 95th percentile of the mean squared error (MSE). The implication is that 95% of the normal data points will have their MSE below the threshold value, and the other 5% above it.

Formula:

$$\text{Threshold} = P_{95}(MSE)$$

Attributes:

$P_{95} \rightarrow$ 95th percentile

Threshold $\rightarrow$ Decision boundary

3.2.8 Anomaly Detection Module

The anomaly detection module identifies whether a particular sequence is normal or abnormal according to the threshold. When the reconstruction error of a particular sequence surpasses the threshold, the sequence is identified as an anomaly. In case the reconstruction error falls below the threshold, then the sequence is regarded as normal.

Decision rule:

$$y = \begin{cases} 1, & \text{if } MSE > \text{Threshold} \\ 0, & \text{if } MSE \leq \text{Threshold} \end{cases}$$

Attributes:

MSE → Reconstruction error

Threshold → Cut-off value

1 → Anomaly

0 → Normal

3.2.9 Interpretation of Results Module

This module deals with the interpretation of results and visual representation of these results. Values for reconstruction errors can be shown on a graph to enable users to analyze any patterns within their own data. High values of reconstruction errors signify an abnormal behavior that needs further investigation, whereas low values suggest normal behavior. Abnormalities might include any strange behavior regarding heart rate or unusual sleeping patterns.

4. IMPLEMENTATION

4.1 Datasets

The dataset contains real-time information about the physical activities and health metrics of a user taken from the Apple Smartwatch. In particular, this dataset is especially suitable for analysing time-series data and detecting any anomalies related to the health monitoring system.

The number of attributes that the dataset contains is quite diverse, which includes heart rate, steps, body mass, height, and timestamps. However, for our purposes, there are only two attributes that need to be analysed: heart rate and steps. The latter will be able to tell enough information about the physiological condition of the person and his activities.

Structure of each entry of the dataset is as follows:

- **Attribute Type** – specifies whether the attribute corresponds to the heart rate or steps
- **Value** – specifies the sensor readings' value
- **Start Date** – specifies the timestamp when the process was recorded
- **End Date** – specifies the timestamp when the process ended

The special thing about time-series structure allows using the dataset in order to analyse user behaviour and apply algorithms of machine learning such as autoencoder to detect any abnormal activity.

4.1.1 Data Preprocessing and Normalization

Data preprocessing makes sure that the wearable dataset is pre-processed and ready for further use. It includes selecting appropriate features like heart rate and steps, discarding irrelevant features, transforming StartDate to datetime format, and sorting data in chronological order.

The numerical transformation of all attributes and handling missing values and outliers is performed next. Then, the data is resampled to hourly frequencies with averaging of heart rate and summation of steps. After that, any missing values in resampled data are filled using forward-fill and backward-fill strategies.

Min-Max normalization is used to rescale all attributes from 0 to 1, ensuring equal weightage during training.

Normalization Equation:

$$X_{norm} = \frac{X - X_{min}}{X_{max} - X_{min}}$$

Attributes:

$X \rightarrow$ Original value

$X_{norm} \rightarrow$ Normalized value

$X_{min} \rightarrow$ Minimum value

$X_{max} \rightarrow$ Maximum value

4.1.2 Training the Autoencoder Model

Training is another essential phase in the proposed system when the prepared time series are used to train an autoencoder to detect anomalies. Its major task is to teach the model to recognize normal behaviour and activity based on the smartwatch data.

i) Generation of Sequences for Time Series

Initially, the prepared dataset is converted into sequences with a fixed length with the help of the sliding windows algorithm. Actually, each sequence is generated to mimic user activities at specific points in time. It is required to identify dependencies and relationships between subsequent sequence members in terms of differences in heart rate levels, number of steps, and energy consumption.

ii) Learning Normal Behaviour Patterns

An autoencoder may be trained exclusively on data about normal behaviour. The rationale behind this decision lies in the fact that the model will have an opportunity to learn only physiological and behavioural patterns. There must not be any anomalies in this phase to provide accurate predictions and tests afterwards.

iii) Autoencoder Architecture

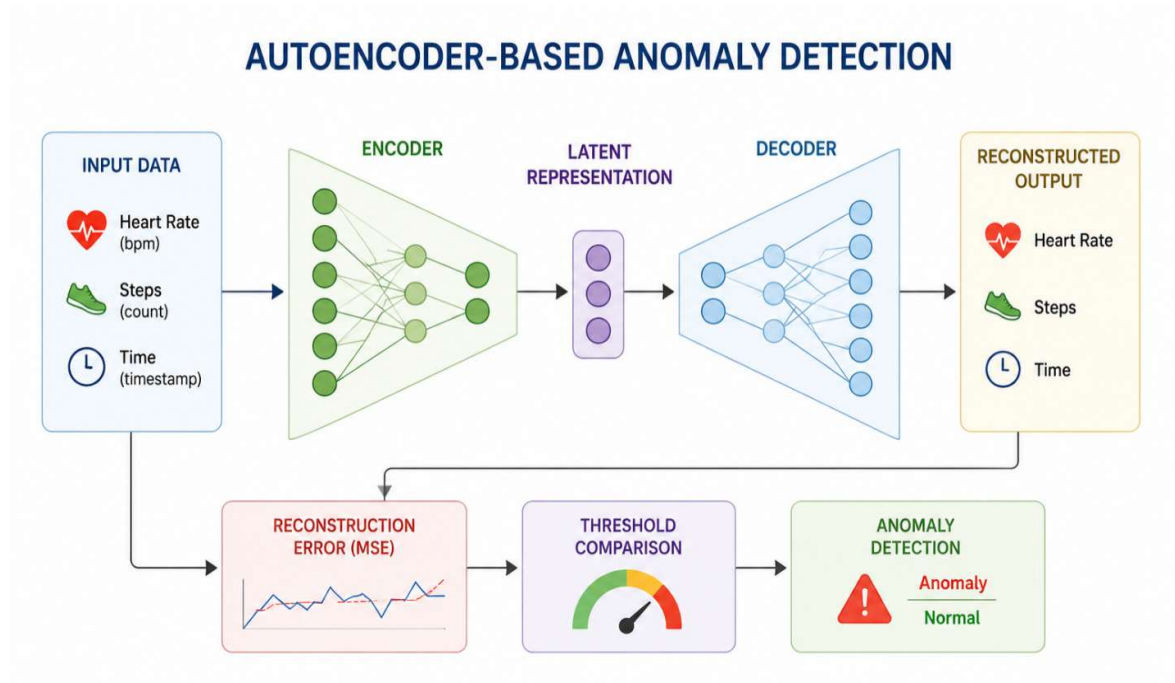


Figure 4.1: Architecture of autoencoder

The proposed architecture as shown in Fig. 4.1 consists of two parts:

- Encoder to transform high dimensional inputs into low-dimensional space in terms of latent variables.
- Decoder to reconstruct the initial input from the latent variables.

iv) Loss Function and Optimization

Learning uses the MSE loss function to compute the difference between the reconstructed output and the input value. This is done by minimizing the reconstruction error needed for accurate reproduction of normal behaviour patterns.

v) Learning Process Behaviour

As learning proceeds, the autoencoder becomes better at learning how to reconstruct normal patterns with reduced reconstruction error. This reflects the learned normal distribution of physiological behaviour patterns in the patient.

vi) Anomaly Detection Sensitivity

Since only the normal behaviour patterns were learned in training, the model does not recognize anomalies and has difficulty reconstructing them resulting in a high level of reconstruction error.

vii) Learning Results Behaviour

Upon completion of training, the model learns to recognize patterns as either normal or anomalies. Reconstruction error is the most important indicator of anomalies in heart rate, physical activity and other physiological data.

4.2 Technologies Used

To implement the described model, it is crucial to utilize certain tools, including those aimed at carrying out the data processing tasks and creating a machine learning model. Specifically, the following technologies have been utilized:

- Python – Programming Language

In this case, it was decided to use the Python programming language for implementing the tasks, and the decision was motivated by the simplicity and clarity of the programming language, which is often used in machine learning projects.

- Pandas and NumPy – Data Processing

These libraries are needed for data processing purposes. Specifically, the Pandas library is required to organize the data properly and filter out any unnecessary observations, while NumPy can assist in calculating some values in order to prepare the data.

- Scikit-learn – Data Scaling

The next step involves using Scikit-learn library, which makes it possible to scale down the data in order to use it with machine learning techniques.

- TensorFlow / Keras – Model Building

In building the autoencoder model, these two technologies will be used. Both platforms enable the building, designing and optimizing of neural network models using loss functions and optimizers.

- Matplotlib – Data and Model Outputs Visualization

The software will be used to visualize time series data. This will involve identifying trends, recognizing patterns and visualizing anomalies in time series data.

- Streamlit – Web Application Building

This framework enables the creation of a dynamic web application for visualizing data results. The technology will help visualize data results including anomalies, and monitor data insights effectively.

- Visual Studio Code (VS Code) – Programming Environment

It will be used for coding and other purposes in data analytics. Some of its advantages include ease in writing codes and debugging among others.

4.3 Autoencoder Model

The autoencoder model is the first algorithm developed as part of this project. This is a simple autoencoder neural network model for analysing health data obtained from a smartwatch, with an aim to classify aberrant behaviour. Data used for training this model is multivariate time-series data received from wearable devices (heart rate, step count, respiratory rate, energy expenditure, heart rate variability (HRV)). Due to the dynamic nature of data received from wearable devices, the model is designed to learn the changing nature of data and classify normal behaviour patterns accordingly.

In the training phase the autoencoder learns a compressed form of the input data, by inputting the data into an encoder network to learn the features and then reconstructing the features with decoder network. The model tries to achieve this through minimizing the difference between the original image and the reconstructed image. The difference is then calculated by means of the Mean Squared Error (MSE) referred to as the reconstruction error. Smaller the reconstruction error, better the model is in learning the underlying distribution while larger the error, worse the deviations are from the normal behaviour.

Overall, the model was trained on data points (sample from the dataset of size 1,280,000) of size 50,000. The model converged smoothly with a final loss of about 0.00237. After training, the model was then used to reconstruct the data, and each point reconstruction error was calculated. The average is about 0.0019 which means most of the data points were reconstructed accurately.

After training, tests were performed on the entire dataset and when a normal behavioural pattern was given as input, the model was able to produce a good reconstruction of the data and the error was very small. When the data given was abnormal, the error grew dramatically. This is how the above was able to detect anomalies.

Observations: The experimental results indicated that the model could represent the normal activities obtained from the wearable data very well. For most of the data the values of the reconstruction error were very close to zero which meant the model learned the data quite well.

When irregularities occurred in the measurements i.e. Sudden high peaks in heart rate, unwanted inactivity where the activity level was far too high, or an inconsistency of physiological parameters, the error in the reconstruction was significantly higher than in normal events. These irregularities made it obvious that they represented an abnormality in the data. Also, the anomalies found were not limited to high or low values they also included behaviour deviations. Another key point to mention is that the models can accommodate to individual behaviour characteristics. Due to the high variation of wearable data between individual users; the learned autoencoder models in the anomaly detection framework can gain high capability in modelling personalized patterns. Observation on the use of features and its effect on model performance has also pointed out that.

Graph Analysis: The graph of reconstruction error, visually illustrates the effectiveness of the model over time. The graphical line for normal data points is continuous and exhibits smooth trend around an error value that is low. Nevertheless, several noticeable peaks are present for some time intervals throughout the graph. These peaks are equally correlated with faults.

These anomaly spikes indicated unusual physiological or behavioural states. For example, if we observed a heart rate spike out of proportion to a lack of physical activity or the duration of the physical activity, or an abnormal pattern in energy expenditure.

The graph makes it easier to interpret results because it enables the user to simply look for irregular trends rather than scan through pages of data to find anomalies. The inclusion of the visualization section in the dashboard adds to the usability and practicality of the system.

Threshold Selection: To identify anomalies, threshold value is needed. For this project, the threshold value is vary based on 95% percentile of the reconstructed error values. According to the experiment result, the value is around 0.00582.

Any test data with a reconstruction error above this value is conventionally classified as an anomaly; the model only allows for 5 percent of test data to be among this grouping; we are therefore using the ‘5th percentile’ of average reconstruction errors for the distribution.

It is also beneficial in that it helped counteract false positives (by not applying any fixed threshold, but rather measurement how the values vary).

Results: A high rate of anomalies was identified by the autoencoder model. Around 5 percent of the data, 64,465 points were identified by the threshold as anomalies. These anomalies were found across abnormal activity, abnormal physiological change and abnormal behaviour.

The model was capable of detecting both the large and small anomalies. It produced low reconstruction error for normal data, yet the abnormal errors were clearly identified.

Efficiency-wise, the model was performing quite well despite the huge size of the dataset. When training on the sampled sub-dataset, the scaled down dataset was much lighter and efficient. The inference process on the full dataset was also very fast and easily scalable. Making it fit for near real time implementation of wearable health-monitoring system, By the way, once we added deterministic seeding into training process, our model was stable and produced rather similar results in different training sessions with selected results being better.

In conclusion, the autoencoder model was a very robust and useful methodology to do anomaly detection for the wearable data. Such model not requiring the usage of labels and the generality and the efficiency that it exhibited would definitely recommend it to be a powerful tool to be used in health analytics.

4.4 Statistical Method

In order to assess the performance of the autoencoder model, a simple statistical method was developed. This is a very simple approach and is often used as a benchmark method for anomaly detection. It uses statistical properties, such as mean and standard deviation, to identify anomalies.

This method calculates the mean for each feature and then set a range by the standard deviation. Any point outside this range can be viewed as an outlier. Like autoencoder, there is no training stage in this method so it is quite quick and simple.

Observations: It is shown from the graph that statistical approach generally has a good performance on extreme value detection. For instance, extreme very high spikes of heart rate and abrupt drops of activity level were successfully detected. But it lacked success in real and complex behavioural models. Wearable data may display a series of smooth variations or slightly different pattern from the mean, which test method has not been successfully identified.

A further limitation of this method is its lack of contextual awareness. For example, when exercising, the heart rate naturally increases, but the statistical method may classify it as abnormal because it deviates from the average. This issue becomes more significant in wearable time-series data, where each data point is dependent on previous states and follows daily behavioural rhythms. However, the statistical methodology assumes no such temporal dependency and treats each data point as independent and isolated from its context.

Thus, it is unable to tell apart deviations in the applied data that are expected, for example the increase in daytime activity, from deviations that are truly anomalous. Its application to time-series data is consequently greatly diminished. Limitations Observed This approach has several limitations. The most important is the large number of false positives. Numerous normal deviations were considered as anomalies since they strayed too far from the mean.

Another weakness is the lack of flexibility. It applies predefined statistical rules rather than adapting to individual user behaviour, so it is inappropriate for individualized health tracking. Furthermore, the method is affected by extreme values. Large values can have an impact on the mean and standard deviation which in turn affects the threshold calculations, resulting in less accurate detection.

Result: The statistical approach was able to detect certain anomalies, especially larger ones. However, it was not able to correctly identify subtle but important anomalies, and it also produced a number of false detections.

All in all, the statistical approach is handy for simple, exploratory processing; however, it is inappropriate for the more advanced and dynamic wearable data.

4.5 Comparison of Algorithms

The difference between the Autoencoder method and the Statistical Method can be seen in the performance, adaptability, and the ability to perform in the real world.

The Autoencoder is a deep-learning, or Neural Network, model that learns on its own through the input of data and finds patterns that define the relationship between the various features of the data. It can also adapt to the changing behaviour of the user over time by adjusting itself continuously. This makes it very well suited for the ever-evolving datasets such as sleep and stress patterns that are inherently dynamic and time-dependent.

On the other hand, the Statistical Method uses well-defined, predetermined rules and threshold limits. The Statistical Method cannot learn from its own data and therefore cannot adjust to the changing nature of the same dataset and can be much less flexible when attempting to deal with complex or changing datasets.

From an analysis perspective, the Autoencoder is much more effective at capturing the time aspect of the data because it can capture the time dependence through the feature representation. The Statistical method does not utilize the time aspect or relationships so its usefulness is limited when analysing a dataset that is time-based or in a sequential manner.

The Autoencoder detects anomalies by evaluating reconstruction error. This property allows the Autoencoder to detect both significant and minor deviations, rather than just strong outliers like many statistical methods do, so it is more effective at detecting anomalies without generating as many false positives or missing more subtle anomalies.

Statistical techniques are easier and cheaper to compute than the Autoencoder, but they may not be suitable for large volumes of data or applications and do not work well with real-time data due to complexity. Rather, the Autoencoder is capable of producing much more accurate and reliable results than statistical methods in most practical scenarios.

Practical Implications

In practice, the Autoencoder would be the best choice for use in wearable health devices to provide accurate insights with minimal false alarms for the user. While the statistical method may provide value for comparison or basic analyses, it is inadequate for complex, advanced applications.

Conclusion of Comparison

The Autoencoder-based methodology provides superior results than its statistical counterpart by providing more flexibility, better sensitivity and greater adaptability. An additional strength of this methodology is that it has the ability to learn from historical information and adapt to changing patterns. As a consequence, it is a more effective and feasible solution for identifying anomalies in health monitoring systems.

5. RESULTS AND DISCUSSIONS

5.1 Results

5.1.1 Login Module

The performance of the system was evaluated based on the outputs generated from the Interactive Dashboard. By using wearable devices, the system shows visual data on how users interact with their wearables along with highlights on any anomalies associated with that usage. Below you'll find the key results in reference to each module utilized.

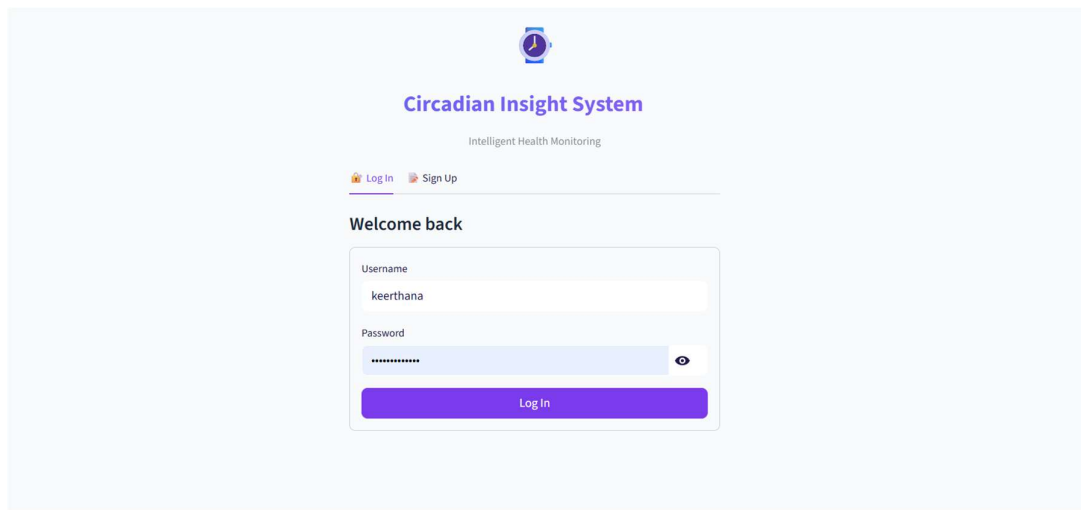


Figure. 5.1: Login Module for Secure Access

Login screen allows users to login to their personalized health dashboard by providing username and password as shown in Fig. 5.1. The data displayed through wearable devices can be restricted to only authorized users.

Main capabilities provided by the login module include:

Ensure user authentication is secure to protect personal health information as illustrated in Fig. 5.1.

Session management in allowing users to access the system during usage.

Determining data that is accessible to logged-in users.

5.1.2 Date Selection and Dashboard Module

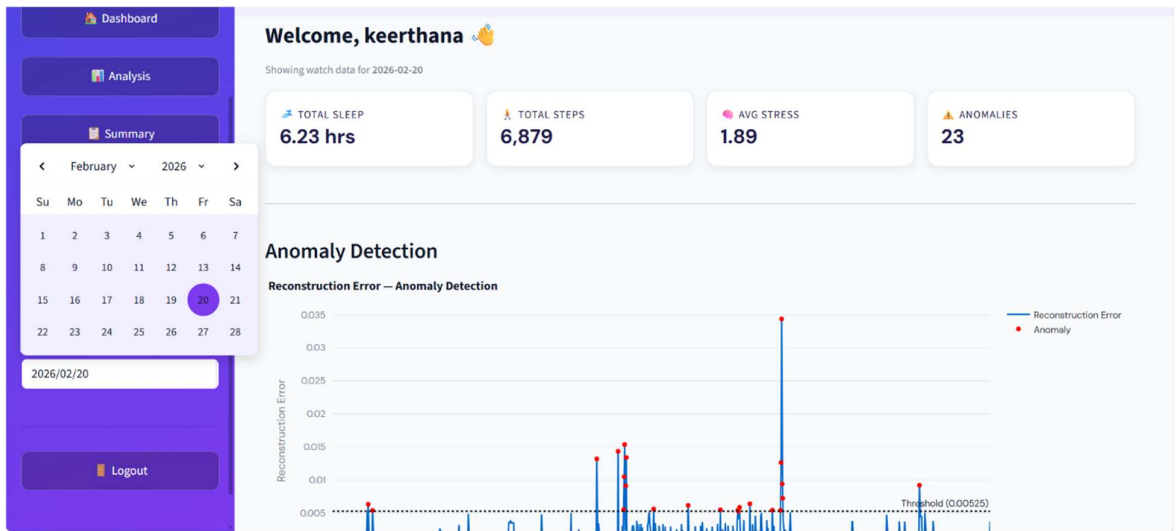


Figure. 5.2: Date Selection with Dashboard

Users can choose a date, and this module will automatically load the corresponding health figures and graphs as shown in Fig. 5.2.

Main features include:

Calendar based date selection for quick navigation between dates as seen in Fig. 5.2.

Real-time updating of dashboard metrics such as sleep, steps, stress and anomalies.

Quick summary view by using cards to better understand daily health.

5.1.3 Anomaly Detection Module

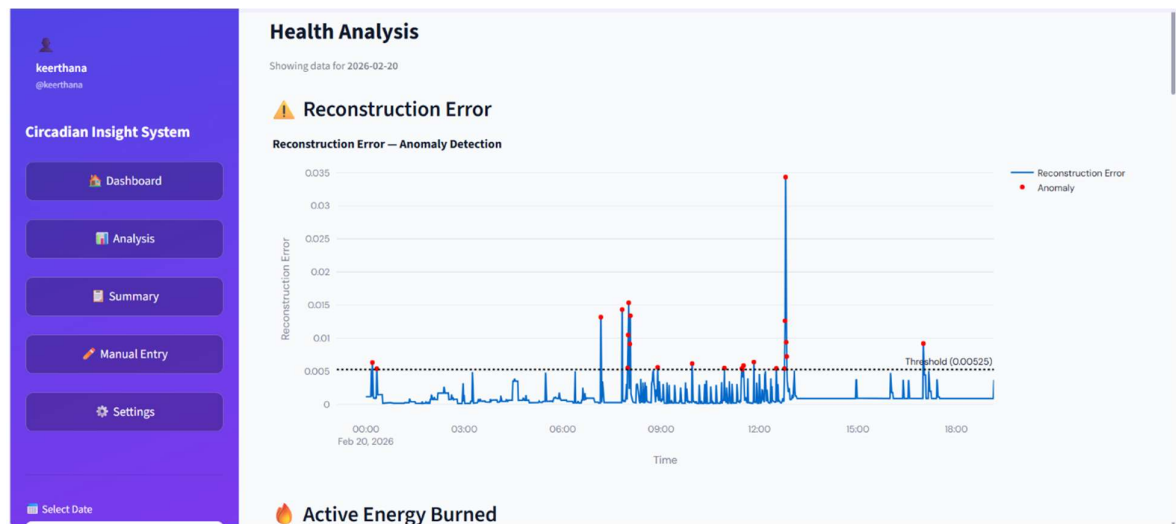


Fig. 5.3: Anomaly Detection Graph

This graph demonstrates the model's detection of anomalies through reconstruction error over time as shown in Fig. 5.3.

Main features include:

Reconstruction error values visualized in time-series.

Threshold line separating normal and abnormal behavior.

Anomaly points are highlighted for easy recognition in Fig. 5.3.

5.1.4 Stress Analysis Module



Fig. 5.4: Stress Analysis

It depicts the fluctuation of stress levels across the day based on physiological information as illustrated in Fig. 5.4.

Main features include:

Tracking continuous stress score over time.

High and low stress identification.

Assists in understanding patterns and variations of daily stress as seen in Fig. 5.4.

5.1.5 Heart Rate Monitoring Module

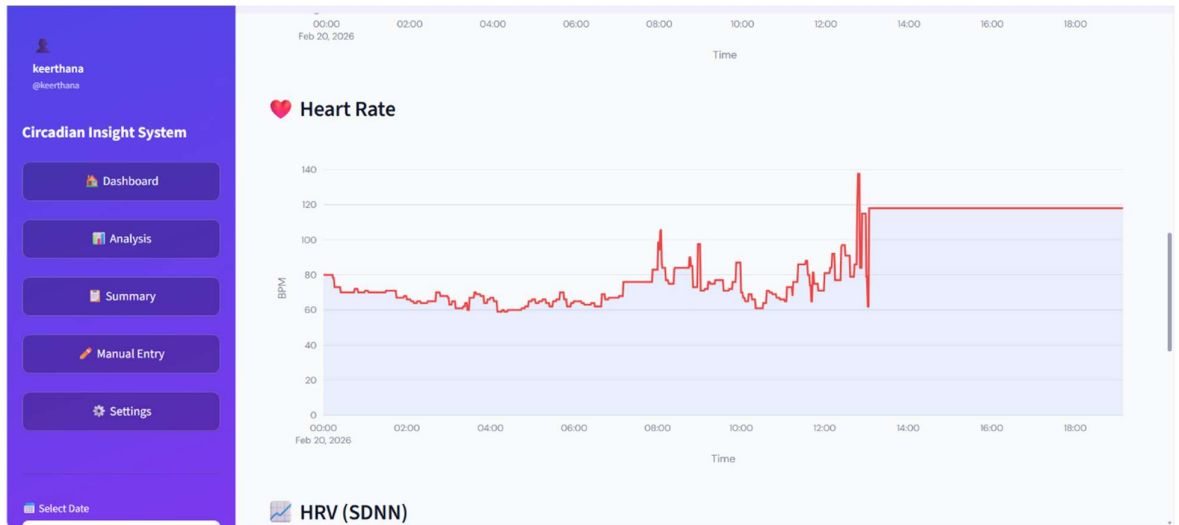


Fig. 5.5: Heart Rate Monitoring

Heart Rate Monitor (Fig. 5.5) Shows the Variation in Heart Rate Over the Course of The Day Based on Resting and Exercising Activities as Shown in (Fig. 5.5) Above

Main Features:

Time-based visualization of heart rate (BPM).

Sudden spikes or irregular patterns are identified.

Useful to relate physical activity with physiological response as observed in Fig. 5.5.

5.1.6 Manual Health Input Module

The screenshot displays the 'Manual Health Check' interface. On the left, a purple sidebar contains the user profile 'keerthana @keerthana' and navigation buttons for 'Dashboard', 'Analysis', 'Summary', 'Manual Entry', and 'Settings'. The main content area is titled 'Manual Health Check' and includes a prompt to 'Enter your health values to get an instant normal/abnormal assessment.' Below this, there are input fields for 'Heart Metrics' (Heart Rate (bpm), HRV - SDNN (ms), Resting Heart Rate (bpm)) and 'Activity' (Step Count, Active Energy (kcal), Basal Energy (kcal)). At the bottom, there is a section for 'Performance & Vitals' with input fields for Physical Effort (0-1), Walking Speed (km/h), Respiratory Rate (br/min), and Sleep (hrs).

Fig. 5.6: Manual Health Input

This module is used when wearable health data is not recorded and allows users to enter health values manually as shown in Fig. 5.6.

Main features include:

Input fields for heart rate, steps, energy, sleep, and other metrics.

Immediate assessment of entered values as normal or abnormal.

Allows continuous monitoring even without device data as illustrated in Fig. 5.6.

5.1.7 Daily Summary and Insights Module

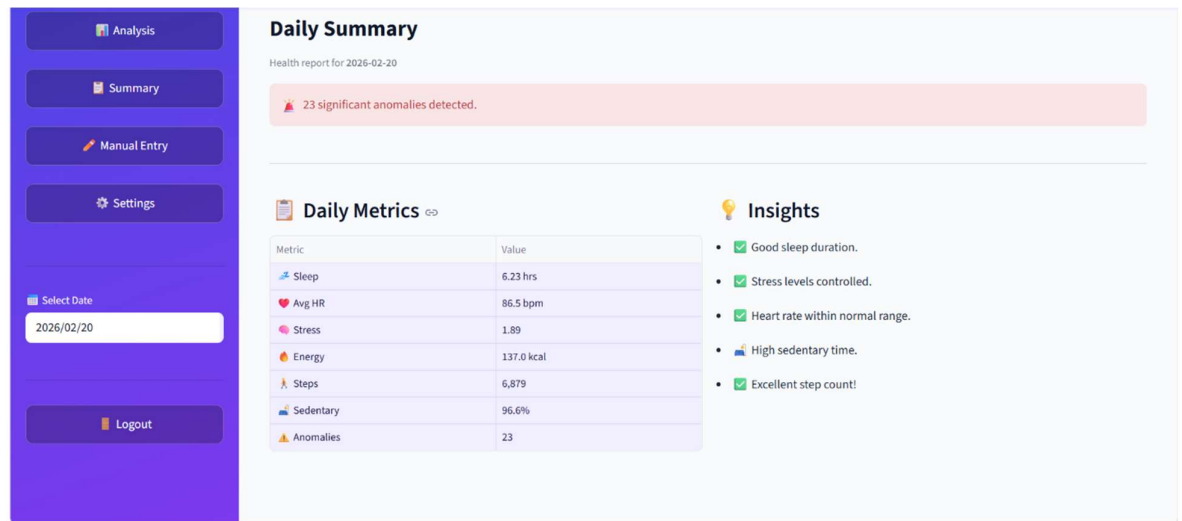


Fig. 5.7: Daily Summary and Insights

Gives a consolidated display of all key health features along with simple derived insights as shown in Fig. 5.7.

Main features include:

Summary table showing important daily indicators.

Highlight of all anomalies detected for that day.

Concise and easy-to-understand findings for quick interpretation as seen in Fig. 5.7.

In summary, the findings from each module demonstrate the ability of the system to analyze wearable data and detect abnormal behaviors as illustrated across Fig. 5.1 to Fig. 5.7. The visualization capabilities provided by the dashboard are user-friendly and aid in interpretation of the data; furthermore, the anomaly detection model performs well in accentuating the abnormality displayed by the data.

The use of multiple health parameters in a combined score is a useful feature for practical activity monitoring applications.

5.2 Performance Measures

The system performance was judged by means of reconstruction error. Because the model uses an autoencoder, traditional accuracy measures are not applicable. Instead, it is measured how accurate the data is reconstructed by the model to identify anomalies. The measure for the calculation of reconstruction error was the Mean Squared Error (MSE). This indicates the squared difference on the mean number of features between the input and its reconstruction.

It is a metric which, when minimized, indicates good reconstruction and normal behaviour and, when high, is an indicator for a bad reconstruction and an anomaly.

Based on the findings, most of the data points had very low reconstruction error values that them being close to zero. This shows that the model has learned the normality in the data. The distribution of the reconstruction error also clearly separated the normal and abnormal data points, which will help to determine anomalies more accurately.

A dynamic thresholding method was employed where the anomaly threshold was determined by the 95% of the reconstruction error values, which gave a value of around 0.00582, above which data points were flagged as anomalies.

Another observation should be pointed out here: afterwards, the same results can be obtained each time we rerun because by turning on deterministic seed. Without setting the initialization random states of the models, the number of anomalies may differ each time. With the seed fixed, all the executions would generate exactly the same result.

The model has also shown quite good generalization ability, because, although the dataset was varied, the error of reconstructed normal behaviour was very small and was very high for abnormal behaviour. That means the model has learnt the behaviour patterns rather than memorise data.

In the speed also the model fared well, though it had to go through more than 12 lakh rows. It trained fast because of the sampling and the prediction was still fast. The entire process can be considered suitable for near real time applications.

In general, the reconstruction error turned out to be an effective metric for anomaly detection on wearable data. In addition, the system maintained a reasonable trade-off between detection accuracy and runtime.

5.3 Comparative Analysis of Results (Tables and Graphs)

Comparison against simple statistic method- For the purpose of analysing the efficacy of the system proposed, it was compared against a simple stochastic system model. The comparison brought out the strengths and weaknesses of the Autoencoder model as shown in Table 5.1.

Table 5.1 Comparison of Algorithms Results

Parameter	Autoencoder Model	Statistical Method
Accuracy	High	Moderate
Time-series handling	Yes	No
Adaptability	High	Low
False Positives	Low	High
Computational Efficiency	Moderate	High
Real-time suitability	Yes	Limited
Pattern Learning	Learns complex patterns	Only detects simple deviations

As shown in Table 5.1, the Autoencoder model achieves higher accuracy and better adaptability than the statistical method. It is capable of handling time-series data effectively and can learn complex behavioral patterns from historical user data. This allows the model to identify subtle anomalies that may not be detected by traditional statistical approaches.

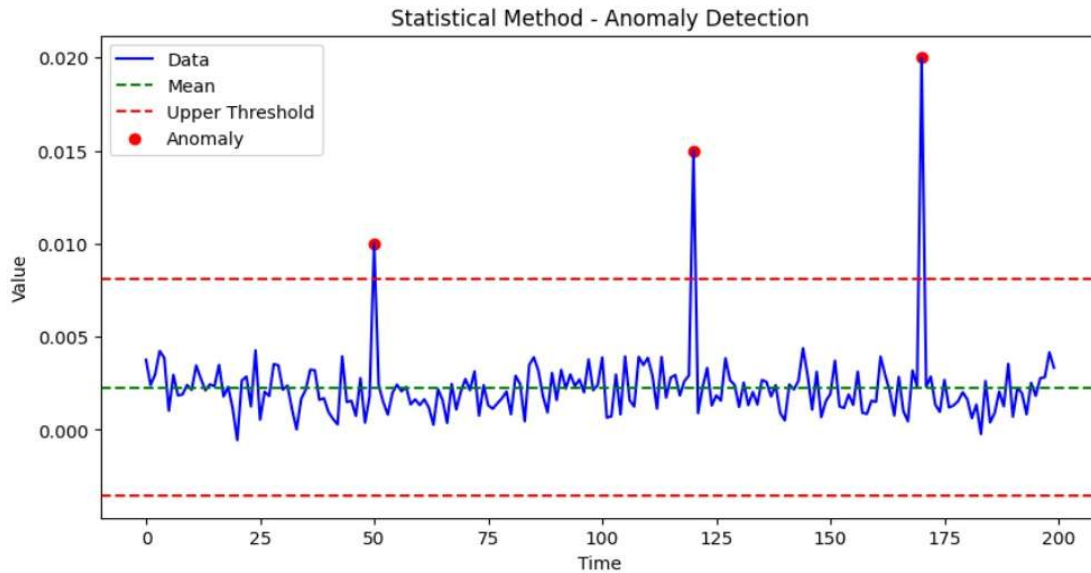


Fig. 5.8: Statistical anomaly detection using mean and standard deviation thresholds

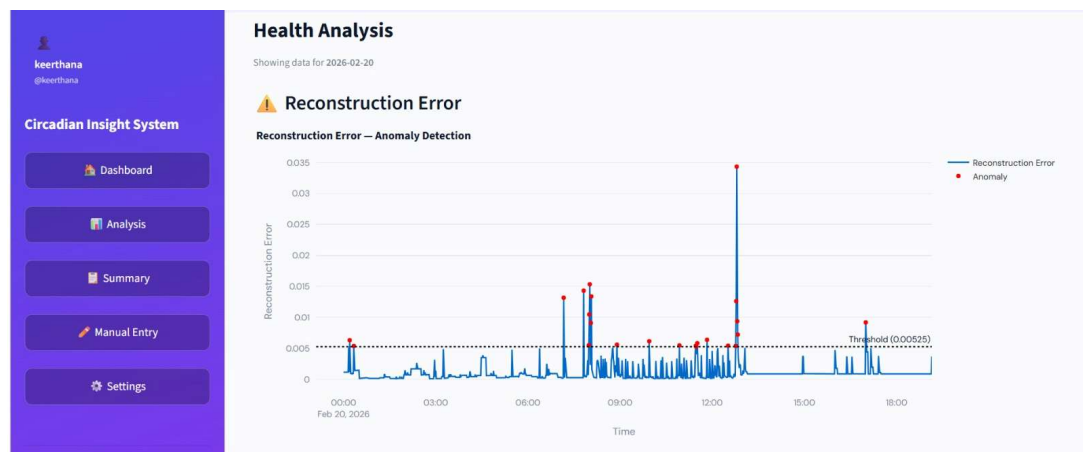


Fig. 5.9: Autoencoder-based anomaly detection using reconstruction error

Detailed Analysis

Table 5.1 shows that the performance of the autoencoder algorithm has been substantially better than traditional statistical methods, as depicted in this table, primarily because of their ability to identify complex and non-linear relations between the different types of physiological data that were input into the model and the way statistical techniques define abnormal behaviour—using a predetermined threshold based off of the mean and standard deviation—as illustrated in Figure 5.8.

Another benefit of using the autoencoder in this research is its application to time-based data, as indicated in Table 5.1. Autoencoders are able to build rolling characteristics and use temporal context to determine how behaviour changes with time. Conversely, the statistical methods use an independent measurement for each data point without considering the time component, in other words, there is no correlation of data points over time.

Statistical methods can only measure large magnitude differences for example large spikes in heart rate and sudden increases in activity as shown in Figure 5.8, and cannot identify more minor variations in behaviour or small, but consistent irregularities around the mean, which created an increased number of false positives.

The autoencoder was successful at identifying both large and small anomalies as shown in fig 5.9. The continuous anomaly scoring of reconstruction error created a more easily identifiable level of severity of all identified anomalies, which produced results that were consistent with the realities of behaviour.

From a graphical analysis perspective, the autoencoder produced acceptable outputs for normal and anomalous data as the autoencoder reconstruction error graph (fig 5.9) produced predictable outputs: smooth trends for normal values, with obvious deltas for anomalous values. Such obvious deltas would also produce outputs that could more easily be identified as abnormal behaviour.

On the other hand, the output from the statistical approach, (fig 5.9) did not produce consistent outputs for normal and anomalous data; thus, the classification of patterns as normal versus anomalous is less clear with this statistical approach. Although the statistical approach is relatively easy to compute, it is very inflexible and cannot accommodate contextual data as outlined in table 5.1; whereas, the autoencoder is much more complex but produce greater levels of reliability and accuracy based upon current methodologies.

Graphical Comparison

The graphical comparison of results as displayed in Table 5.1 clearly illustrate the results of Table 5.1 and the results of graphing the reconstructed error of the autoencoder where the reconstructed error is displayed in Figure 5.9. The reconstruction

errors of the normal data set produced a stable pattern with spikes caused by anomalous events; this makes it easy to interpret the difference between the errors of reconstructed data and the actual data.

While the mean and standard deviation of the raw data generated from the statistical analysis of the wearable time-series data using the statistical methodology are useful for identifying only the extreme outliers, you can also precisely identify large spikes. You will not be able to correlate smooth lines before and after large spikes; in addition, the produced image is not as consistent, nor is the boundary from abnormal to normal as readily apparent as it is in the graph of the autoencoder reconstructed data in Figure 5.9.

In summary, the visual comparison of the results we obtained using an autoencoder to identify anomalies in time-series wearable data is superior to the results we obtained using traditional statistical modelling methods, as indicated in Table 5.1, Figure 5.8 and Figure 5.9.

Discussion

In summary, it can be concluded that the autoencoder method is superior to the other methods discussed in this chapter (see Table 5.1 and Figure 5.9). It outperforms previous methods and approaches because it learns from actual data and optimizes its accuracy, versatility, and consistency directly as it learns from the data collected.

Statistical methods such as those found in (see Figure 5.8) are helpful in establishing a simple reference point for comparison; however, they do not apply to dynamic and more complex data sets where the behavioural aspect of the user's data can change over time.

Overall, the system operates very well, detects anomalies in the user's data, and gives useful information about the user's activity.

6. CONCLUSIONS AND FUTURE ENHANCEMENTS

6.1 Conclusions

The project was able to develop an anomaly detection system based on the analysis of smartwatch-based time series input (heart rate, steps, energy expenditure, HRV). Data pre-processing involves dealing with missing values, noise, and timestamps and the structuring of data for the analysis process.

In this context, the lightweight autoencoder architecture (15→7→15) was applied, providing minimal reconstruction loss (0.0023). The detection of anomalies was achieved based on the MSE reconstruction error with a threshold value equal to the 95th percentile (0.0058).

The developed system works effectively with big amounts of data and without the need for labelled data, which makes it applicable for further use in the real-world environment. At the same time, compared to traditional statistical techniques, the results of detection are better and the dashboards make the interface easier to understand.

6.2 Future Enhancements

The proposed system can be extended by including additional physiological parameters (SpO₂, ECG, and sleep data) to improve precision. Real-time monitoring pipeline can provide fast processing.

Further improvements will consist in using more powerful autoencoders (LSTM), personalising the models, as well as storing data in the cloud and providing notification capabilities.

REFERENCES

- [1] A. Géron, "Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow," O'Reilly, 2019.
- [2] C. C. Aggarwal, "Outlier Analysis," 2nd ed., Springer, 2017.
- [3] Centers for Disease Control and Prevention (CDC), Physical Activity and Heart Rate Guidelines, Centers for Disease Control and Prevention, Available: <https://www.cdc.gov>
- [4] I. Goodfellow, Y. Bengio, and A. Courville, "Deep Learning," MIT Press, 2016.
- [5] MDPI Sensors Journal, Wearable Sensor Data Analysis Using Machine Learning Techniques, MDPI, Available: <https://www.mdpi.com/1424-8220/21/9/3235>
- [6] National Institutes of Health (NIH), Wearable Health Monitoring and Biomedical Research, National Institutes of Health, Available: <https://www.nih.gov>
- [7] National Library of Medicine (NLM), Circadian Rhythm and Physiological Monitoring Research, National Institutes of Health, Available: <https://www.ncbi.nlm.nih.gov>
- [8] National Sleep Foundation, Sleep Health and Sleep Duration Recommendations, National Sleep Foundation, Available: <https://www.sleepfoundation.org>
- [9] PLOS ONE Journal, Machine Learning-Based Circadian Rhythm and Health Analysis Using Wearable Data, Public Library of Science, Available: <https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0279305>
- [10] PubMed Central, Wearable-Based Health Monitoring and Circadian Rhythm Detection Study, National Library of Medicine, Available: <https://pubmed.ncbi.nlm.nih.gov/39163188/>
- [11] World Health Organization (WHO), Global Health Observatory Data Repository, World Health Organization, Available: <https://www.who.int>

GLOSSARY

Autoencoder

A neural network models designed to learn compressed representations of data and to reconstruct its original form. In this project it is used mainly for anomaly detection.

Anomaly

An example of a data point that is quite far away from the normal behaviour. In this project, anomalies mean unusual health or activity levels.

Time-Series Data

Data collected over intervals of time. In this particular case, the data from the watch is accumulated over a period of time so data like heart rate, number of steps etc is taken in continuous flow from the watch.

Reconstruction Error

Difference between the original input data and the output of the autoencoder, used to identify anomalies.

Mean Squared Error (MSE)

A measurement of the reconstruction error over a data set by taking the average of the squared differences between actual and predicted values.

Normalization

The method of rescaling data to a fixed range (most commonly 0 to 1) where prior to normalization, it is often desirable to do some preparation steps.

Threshold

This value is used to distinguish between normal and abnormal data, in this project the percentile value 95th of has been used. 95th percentile is used as cutoff.

Feature Engineering

Extraction of new meaningful features (stress score, sedentary index and recovery index, etc) from raw data.

Sedentary Index

A feature derived that indicates times of inactivity.

Stress Score

A derived feature from HR and HRV which attempts to quantify the amount of stress a person is undergoing.

Recovery Index

A feature which shows the level of body recovery according to physiological signals.

APPENDIX

Appendix A: Dataset Structure

Dataset system employed: CSV-based Apple Watch Health Export

Tables/Collections:

Health Data Table: contains time-series health measures such as heart rate, steps, respiratory rate, energy burned (active and basal), walking speed, energy expenditure, and heart rate variability with respective timestamps

Sleep Data Table: entries from the sleep section of the dataset. Sleep segments recorded and total sleep per day calculated from the smartwatch data set

Processed Data Table: has cleaned and feature engineered data collected from the cell phones with these additional derived metrics (stress score, sedentary index, recovery index, rolling averages)

Final Data Table: processed data with autoencoder's output reconstruction error provided by the autoencoder model and anomaly label.

Appendix B: Technology Stack

Frontend: Streamlit (Python interactive dashboard)

Backend: Python

Data Processing: Pandas, NumPy

Machine Learning Framework: TensorFlow / Keras

Machine Learning Techniques: Autoencoder-based anomaly detection

Visualization Libraries: Matplotlib, Plotly

Development Environment: Python Virtual Environment

Versioning: GitHub

Appendix C: Screenshot Samples

Dashboard Home Interface (Summary of health indicators)

Reconstruction error graph with anomaly spikes

Daily Sleep Monitoring Visualization

Stress Score and Activity Trends

Anomaly Detection Output Panel

Appendix D: Dataset Information

Dataset Employed: Apple_watch_data.csv

Size of data: Approximately 1.28 million after finding missing values and data cleansing.

Variables Included:

Heart Rate

Step Count

Respiratory Rate

Active Energy Burned

Basal Energy Burned

Walking Speed

Physical Effort

Heart Rate Variability (SDNN)

Resting Heart Rate

Timestamp

Derived Features:

Stress Score

Sedentary Index

Recovery Index

Rolling Averages (a form of window based smoothing)

Data Type: Real life wearable health data collected from sensors.

Appendix E: Algorithm and Model Description

Primary Model: Autoencoder Neural Network

Utilise for detections of anomalies non-supervises at learning normal behaviour and identifying anomalies.

Model Architecture:

Input Layer (15 features)

Hidden Layers for dimensionality reduction

Bottleneck Layer (compressed rep)

Decoder Layer for reconstruction

Output Layer (same dimension as input)

Training Method:

Model trained on sampled dataset (~50,000 rows)

Optimization with MSE loss function

Anomaly Detection Technique:

Reconstruction error for each point.

Increased error implies deviation from pattern

Thresholding Method:

The dynamic threshold is set as, 95th percentile of the reconstruction error

All ratios above threshold identified as anomalies

Appendix F: Performance Metrics Used

Reconstruction Error (MSE): It is used for quantifying the difference between the input and the reconstructed output.

Mean Reconstruction Value: Shows performance of model on normal data.

95th Percentile Threshold: Can be used as a cut-off point separating the normal and the abnormal data.

Anomaly Ratio: percentage of anomalies detected (~5% of data set)

Training Loss: Final loss value after model converging (~0.00237)

Processing Efficiency: Able to process large-scale time-series data

Appendix G: Possible Future Developments in the System

Inclusion of other health variables such as SpO₂, sleep stages, stress state, etc.

LSTM implementation for higher time-series learning.

The evolution of real-time mobile health monitoring programs

Deployment to the cloud for storing and processing huge amounts of data

Individualized anomaly detection models for each user

Real-time alert system integration for anomalies notification

Appendix H: More Information About the System

Engineered to handle auto-scalable, large scale time series wearable data

Utilizes unsupervised learning so it does not require training data to be labelled

Includes feature engineering for enhanced behavioural insights

Interactive dashboard is provided for visualization and interpretation

Ensures reproducibility when training model by setting random seed

Indicates that it is acceptable for real-time and near real-time health monitoring applications